

Lecture 1: Selected Topics in HTML5 and CSS3

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Factors that make a good web application

- Looks clean and easy to use?
- Responsive on mobile?
- Fast?
- Accessible?
- SEO-friendly?
- Secure?
- Easy to maintain?
- Works across browsers?
- Uses semantic HTML?
- Supports dynamic content?

How to make the web application more responsive?

Responsive: website adapts seamlessly to different screen sizes – desktop, mobile or tablet screens

- Flexible layouts such as CSS Flexbox or CSS Grid box
- Media queries
- responsive units

What can you to make your web application faster?

- Optimize resources-
 - compress HTML, CSS and JS, images (WebP)
 - Reduce network load- use caching effectively
- Load intelligently
 - Lazy loading for images and videos
 - Code splitting and dynamic import for large apps
 - Defer non-critical JS
- Optimize the backend
 - Optimize database queries
 - Use server side caching
- Measure and monitor
 - Use browser dev tools/ Lighthouse
 - Core web vitals
 - Identify and fix bottlenescks

How to make it more secure?

- Protects user data and prevents unauthorized access.
- Uses HTTPS/TLS, authentication, authorization, and secure session management.
- Protects against threats such as XSS, SQL Injection, and CSRF.
- Validates and sanitizes user input.

How to make it SEO optimized?

- SEO: Makes content easier for search engines to discover, crawl, and index.
- Uses meaningful titles, headings, meta descriptions, URLs, and structured content.
- Optimizes page speed, mobile experience, and content quality.
- On page and Off page SEO methods

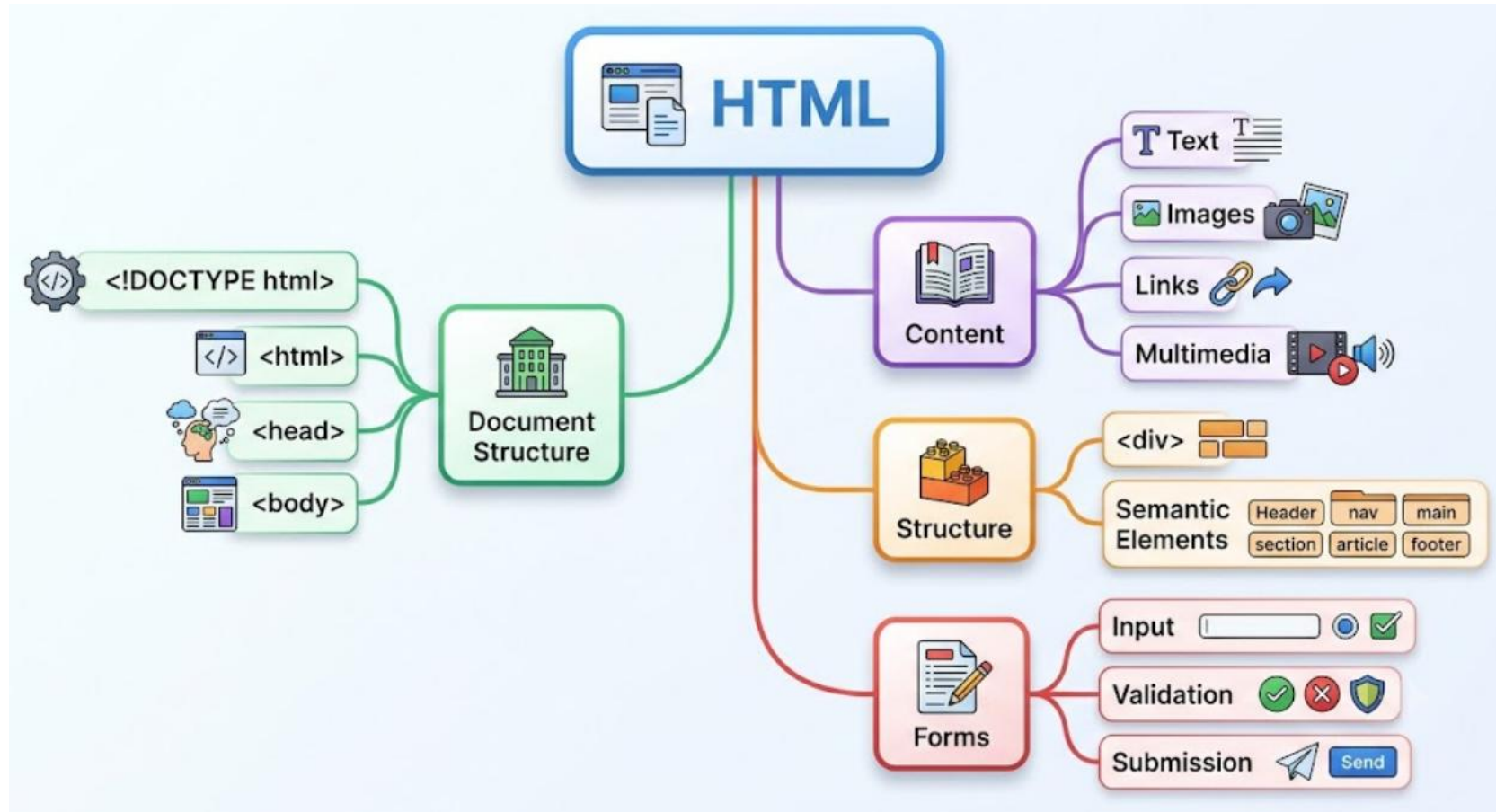
How to make is accessible?

- Accessible: Can be used by people with different abilities and disabilities.
- Follows accessibility guidelines such as WCAG.
 - Web Content Accessibility Guidelines (WCAG) are the internationally recognized standards for making digital content-such as websites, mobile apps, and documents accessible to people with disabilities.
 - Developed by the World Wide Web Consortium (W3C), they outline specific success criteria to ensure content is usable for everyone
- The 4 pillars of accessibility (POUR)
 - Perceivable: UI and information to be presented in a way that users with disability can easily perceive- sufficient color contrast, and alternative text for images.
 - Operable: Users must be able to navigate and interact with all features using various inputs, not just a mouse (Supports keyboard navigation, screen readers,
 - Understandable: Information and interface operations must be clear, predictable, and simple to comprehend Uses semantic HTML to improve accessibility and understanding of page structure.
 - Robust: Content must be reliable enough to work well with current and future assistive technologies, like screen readers.

Accessibility Conformance levels:

- Each WCAG guideline contains testable success criteria rated at three levels of compliance:
 - Level A: The most basic, minimum level of web accessibility.
 - Level AA: The most common standard targeted by businesses and legal frameworks (like the ADA).
 - Level AAA: The highest and strictest level of accessibility.

Overview of HTML5 Elements



Why did HTML5 introduce semantic elements when we could build everything using `<div>`?"

HTML5: semantic markup

- "what role does that HTML element have?" rather than "what does it look like?"
- Existing semantic markup:
 - Consider two ways of writing a heading in html

```
• <h1>This is a top level heading</h1>  
• <span style="font-size: 32px; margin: 21px 0;">Is this a top level heading?</span>
```

- First way uses the semantic tag and thus is a more appropriate way of markup!
- Other examples of semantic elements: <table> <form> <h1>
 - Examples of non-semantic elements: <div> and

Semantic markup benefits

- Semantic markup is self-describing and uses the correct HTML elements for the correct job
- Search engine optimization (SEO): Search engines will consider its contents as important keywords to influence the page's search rankings
- Screen readers can use it as a signpost to help visually impaired users navigate a page
- Ease of collaborative development/maintenance
 - Finding blocks of meaningful code is significantly easier than searching through endless divs
 - Suggests to the developer the type of data that will be populated

HTML5 Structural Elements

- How to divide page into regions and sections?
- HTML4
 - We could markup using classes and IDs, eg.

```
<ul class="nav">  
<div class="sidebar">
```

```
<ul class="nnn">  
<div class="sssss">
```

- Both are same
- Extra semantics not accessible for users
- Browsers just ignore them except for styling

Popular class names and their HTML5 equivalents

Table 3-1. Popular Class Names and Their Approximate HTML5 Element Equivalents

Class Name	HTML 5 Element
Footer	<footer>
Menu	<menu>
Title, header, possibly top	<header>
Small, copyright, smalltext	<small>
Text, content, main, body	<article>
Nav	<nav>
Search	<input type="search">
Date	<time>

Typical HTML4	Typical HTML5
<div id="header">	<header>
<div id="menu">	<nav>
<div id="content">	<section>
<div class="article">	<article>
<div id="footer">	<footer>

HTML4: Extra semantics not accessible for users
Browsers just ignore them except for styling

New Structural Elements

- `<section>`
 - `<article>`
 - `<nav>`
 - `<aside>`
 - `<header>`
 - `<footer>`
- ▶ Referred to as structural elements
 - ▶ Replacement for common ad-hoc semantics for parts of the page
 - ▶ Standardised and *media-independent* semantics
 - ▶ it should work across different platforms, devices and media

Sectioning content elements:

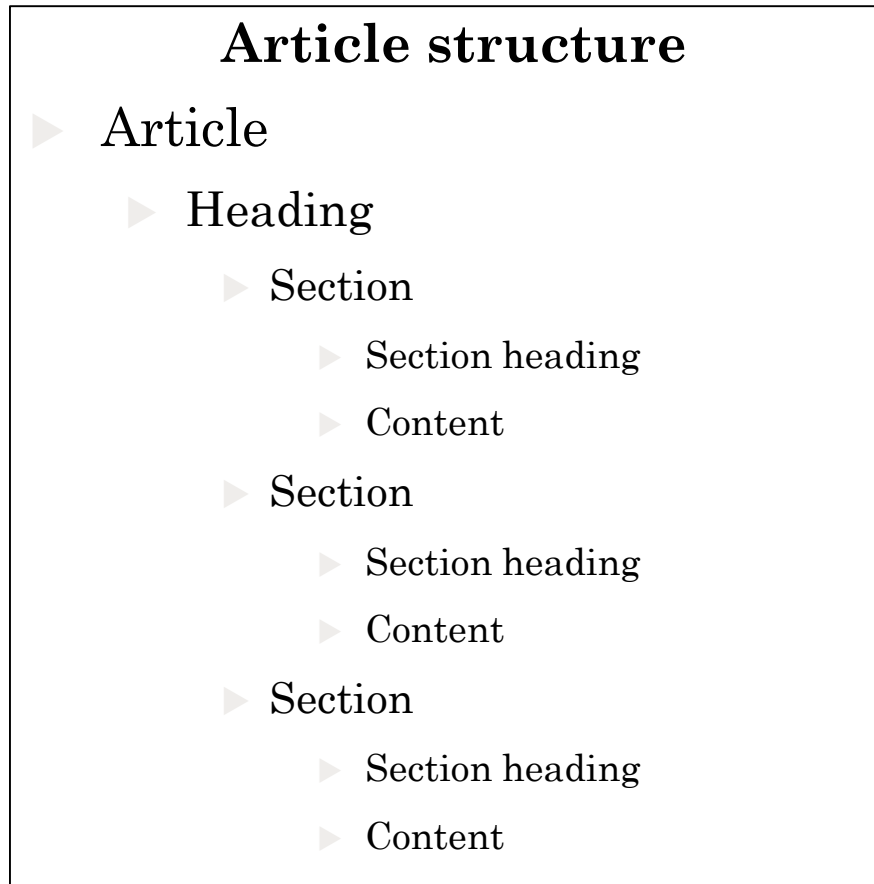
<section> and <article>, <nav>, <aside>

Structural Building Blocks – `<div>`, `<section>` and `<article>`

- **`<div>`:**
 - The generic container. It's a flow content element with no additional semantic meaning.
- **`<section>`:**
 - A generic document or application section, almost always with a heading (maybe in a `<header>`) and sometimes a `<footer>`, too.
 - It's a chunk of *related content*,
 - e.,g. subsection of a long article, a major part of the page (like the news section on the home page), or a page in a web application's tabbed interface
- **`<article>`:**
 - An *independent* section of a document or site. This means it should be able to stand alone and still make sense if you encounter it somewhere else
 - Examples include a weblog article, a forum post, or a comment.
 - Like `<section>`, these should have a heading and may have a header and/or footer.

Example: Sections within an article

A long article with subsections (like a thesis)



Headings: <header> and <footer>

- **<header>**: Used for **introductory and navigational content of a sectioning element**.
 - includes the heading (an <h1>–<h6> element or an <hgroup> element), but can also contain other content, such as a <nav> element or navigational links, a table of contents, a search form, or any relevant logos.
 - It **can't** contain <footer> or another <header>. <header>'s defining purpose is to contain *introductory or navigational aids*, rather than define their placement (headers are usually at the top of a section, but **not necessarily**).
- **<footer>**: Used for **additional information about the content**, such as who wrote it, links to related documents, copyright data, a link to the top of the page, etc., and usually **appears at the end of the content**. Like <header>, it can't contain <header> or <footer> elements.
- <header> and <footer> apply to the current sectioning element. If that element is <body>, then the <header> or <footer> does apply to the whole page. But if it's another sectioning element, then <header> or <footer> only applies to that element.

- An article with one heading

This has a heading `<h1>` only. No `<hgroup>` or `<header>` is needed.

```
<article>
  <h1>Article heading</h1>
  <p>Content...</p>
</article>
```

- An article `<header>` with heading and metadata

Now we wrap a heading and metadata in `<header>`.

```
<article>
  <header>
    <h1>Article heading</h1>
    <p>(<time datetime="2009-07-13">13th July, 2009</time>)</p>
  </header>
  <p>Content...</p>
</article>
```

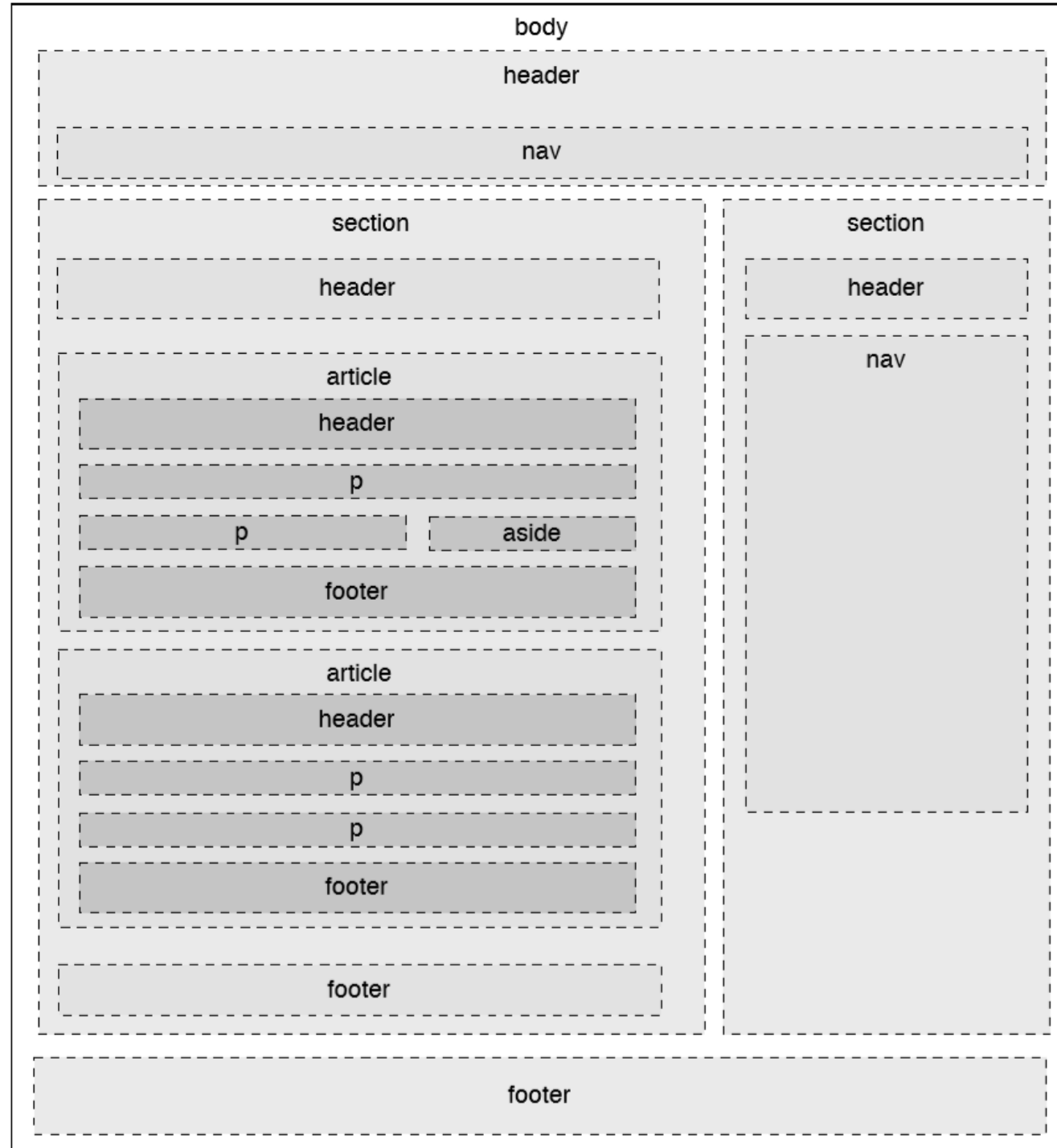
<nav>

- Section of navigational links
 - to other pages (site navigation)
 - Or to sections on the same page (eg. ToC)
- Pre-HTML5 way of creating navigation using <ul class="nav">
- <nav> allows explicit markup for navigation
 - Accessibility benefits – skip navigation and go straight to the content

```
<nav>
  <h2>Main navigation</h2>
  <ul>
    <li><a href="/">Home</a></li>
    <li><a href="/blog/">Weblog</a></li>
    <li><a href="/about/">About</a></li>
    <li><a href="/contact/">Contact</a></li>
  </ul>
</nav>
```

<aside>

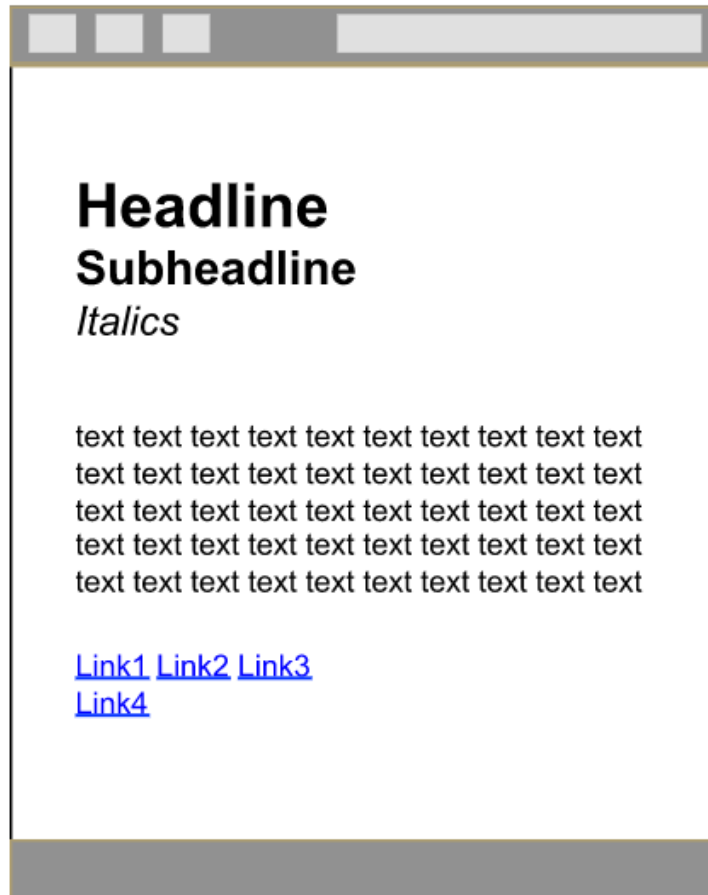
- It's basically a side note
- Content that is additional to but not essential for understanding the main content.
- Could also be used for reserving advertisement space



Structured data

What browsers see vs. what humans see.

Can we make content **machine-readable** so that browsers, scripts, or robots see more of what we see?



Structured Data

- Data organized in a predictable format for easy interpretation by machines.
- Enhances search engines' understanding of web content.
- Why is it important?
 - Enables rich search results (snippets, carousels, knowledge panels).
 - Improves SEO visibility and indexing.

Vocabularies

- Vocabulary: Set of standard terms to describe things (people, products, events).
- [Schema.org](http://schema.org):
 - A collaborative vocabulary supported by Google, Microsoft, Yahoo, and Yandex.
 - Provides schemas (types, properties) to mark up content.
 - Used with microdata, RDFa, or JSON-LD.
- These vocabularies cover entities, relationships between entities and actions
- semantic information in the content can be conveyed in a way search engines can understand
- Structured data testing tool/ Rich results test available

JSON for linking data (JSON-LD)

- Based on Javascript object notation (JSON)
- JSON-LD is a scripting language that allows publishers to communicate important information to search engines.
- creates a network of standards-based, machine-readable data across Web sites.
- **JSON injects metadata in the head section.** This is used as a data block added as a script element. No need to tweak actual HTML data.
- Loads faster, better indexing in search engines
- It is the choice of search engines like Google and Yahoo
- JSON-LD structured data influences how Google will create rich snippets for your URLs.

Rich snippet

- Rich snippets are enhanced search results shown on search engines
- Generated when a webpage includes structured data markup (like JSON-LD, Microdata, or RDFa) that helps search engines understand the page content better.
- Importance:
 - Make your listing stand out in search results
 - Can improve click-through rates (CTR).



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